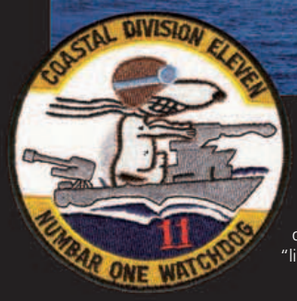


Missile warfare
The Nimitz-class aircraft carrier USS *Abraham Lincoln* fires a RIM-7 Sea Sparrow surface-to-air missile. Missiles have largely superseded guns as the primary air and surface defense on board modern warships.



Vietnam patch
America's war in Vietnam saw an increasing focus on coastal and riverine warfare—and the formation of naval units dedicated to fighting in these "littoral" environments.

world but only just capable of independent action—as it showed in the Falklands War in 1981. The end of the Cold War in the late 1980s left the US Navy with an absolute predominance over any potential enemies, though with plenty of work for its aircraft and missiles to perform in support of global interventions, especially in the troubled Middle East.

CHANGING NAVIES

Battleships for many years continued to function in a shore bombardment role, but the capital ships of the postwar era were unquestionably carriers and submarines. Carriers grew in size to accommodate jet aircraft—although jump jets and helicopters enabled all kinds of smaller warships to mount their own aircraft—and were equipped with nuclear engines that gave them almost unlimited operational range.

Genuine submarines, in the sense of a vessel capable of sustained operations underwater, were brought into service by Germany at the end of World War II, but it was nuclear power that really made submarines come into their own. Nuclear submarines, much bigger than their predecessors, could travel under the sea almost indefinitely. They diverged into two types: those armed with ballistic missiles that formed part of the "nuclear deterrent" and hunter-killer submarines, chiefly to be used to hunt an enemy's submarine forces.

The demise of the big guns did not mean that the only valid surface warships were carriers. Frigates, reintroduced during World War II, and destroyers continued to fulfill a vital role as platforms for anti-submarine, anti-ship, and anti-aircraft missiles, as well as mounting nuclear and non-nuclear land attack missiles. The predominance of missiles was such that by the 1970s some warships were being built with no guns at all—but this proved a momentary aberration. The use of missiles, with their guidance systems and the countermeasures they required, was necessarily linked to massively sophisticated systems of communications and control. Combat was in the hands of men or women sitting in front of computer screens in control rooms who would never see the enemy they fought. By the 1991 Gulf War, shore bombardment was being conducted at ranges of hundreds of miles not only by naval aircraft but also by Tomahawk cruise missiles.

THE 21ST-CENTURY NAVY

By the 21st century it was fair to say that, in terms of world power, the US Navy was the only force that really counted. The navy that took part in attacks on Afghanistan and Iraq was a refined instrument of global power-projection. Its crews, ratings as well as officers, consisted mostly of educated people—women as well as men from 1993—exercising some special skill. Many were operating electronic gadgetry related to sensors and guidance systems, control and communications. Manual labor formed only a relatively small part of life on board any warship. Even on submarines, crew experienced conditions—in terms of food, comfort, and medical care—far more civilized than any of their historical predecessors. What role this superbly trained and equipped navy would find to play in the context of the "War on Terror" remained open to question.



Missile keys
The set of keys used to arm the Polaris nuclear missile on board the ballistic missile submarine HMS *Repulse*.

1973
US troops complete withdrawl from Vietnam

1973
Israeli missile boats sink several Syrian ships near Latakia in first engagement between boats using surface-to-surface missiles and electronic countermeasures

1982
Falklands War between Argentina and UK: sinking of a British destroyer by Exocet missile illustrates the vulnerability of surface craft to anti-ship missiles

1988
US Navy attacks Iranian oil platforms in the Gulf in retaliation for Iranian minelaying

1990
Iraqi invasion of Kuwait sparks the First Gulf War

1991
Collapse of Soviet Union leaves USA as the world's only naval superpower

2001
Muslim extremists destroy World Trade Center in New York

2001
US invasion of Afghanistan begins, supported by carrier groups in the Gulf

2003
US invasion of Iraq opens the Second Gulf War

2006
USS *Freedom* launched, the first ship in US Navy's next generation of Littoral Combat Ships

1970 1980 1990 2000 2010



Soviet submarine
An Akula-class nuclear-powered attack submarine, deployed by the Soviet Navy in 1986.